



SCHOOL OF COMPUTING AND ENGINEERING SCIENCES
BACHELOR OF SCIENCE IN INFORMATICS AND COMPUTER SCIENCE
COURSE OUTLINE

ICS 4101: Informatics and Computer Science Project II

Lecturer: Khakata, E. N. (PhD.) **Email:** egathenya@strathmore.edu

Consultation: in Class/ on Email / Module Leader **Module Leader:** **TBD**

Semester Dates:

Sem I: 7th April – 20th July 2026 & Sem II: 17th August – 1st December 2026

Purpose: To carry out original works that applies the theoretical and practical knowledge in Informatics and Computer Science.

Intended Learning Outcomes:

At the end of this course, the learner should be able to

1. Apply technical writing and proper documentation.
2. Apply project management and software construction.
3. Make presentations of their Project.

Contact Hours: 2 Semesters

Prerequisite: ICS 3205: Research Methodology and Technical Writing

Lecture time:

ICS 4D: Tuesdays 11:15am – 1:15pm STMB F1-02 & Wednesdays 12:15 – 2:15pm

ICS 4E: Wednesdays 8:15am – 10:15am STMB F5-01 & Fridays 8:15 – 10:15am

Content, Outcomes and Activities

- This is a project in which the candidate develops an application/a system/ a solution that can be implemented, using the techniques introduced in the earlier courses.
- Each project will be supervised by a member of the teaching staff and will run for two semesters. The 15 supervisors are AKhajira, Dr. DOWuor, Prof. VOMwenga, Dr. EKhakata, Dr. SObonyo, AVikiru, DMachanje, EManyasi, EMenya, EOlang', JGikera, JKirui, KOMondi, LBonyo, TTabulu, SMwongela, SMonthe,
- Students will be expected to attend a series of seminars during the class sessions in order to give feedback regarding the different milestones achieved.
- They will also be expected to be examined during two oral presentations of their projects within a semester so as to demonstrate their progress in the project.
- The deliverables are the developed application/ system/ solution and its related documentation.
- The entire project will be checked and marked for advanced programming, human computer interface and design, advanced database techniques, networking, mobile programming, distributed systems and modelling skills that were acquired in Stage II.

Semester 1: 7th April 2025 to 18th July 2025

Week/ Dates	Topic	Intended Learning Outcomes	Hybrid/Hyflex Learning Activities		Assessment
			Face to Face/On-Campus	Online	
Week 1-2	Project Idea & Concept Note (7 th April – 18 th April)	At the end of this topic, students will be able to: a) Come up with an idea to solve an Informatics/Computer Science related problem that has relevant application to the immediate society or environment b) Write a half page Concept Note that gives a brief explanation of the idea that you have come up with. c) Submit the concept note.	• Idea brainstorming. • Conceptualization of the thought of ideas.	• Progress check • Discussions	
Week 3-4	Project Supervision (21 st April – 1 st May)	At the end of this topic, students will be able to: a) Communicate with their allocated Supervisor. b) Contact their supervisor, discuss the idea and see the way forward. c) Note down on the ICS Project Supervision Sheet the supervision meetings held in the year.	• Refining of the Project Idea • Supervisor assignments.	• Progress check • Discussions	

Week 5-8	Proposal Document (4th May – 12th June)	At the end of this topic, Students will be able to: a) Draft a proposal document as discussed in class. b) Review the document with the supervisor. c) Make corrections on the document as advised by the Supervisor. d) Submit the corrected copy for Examination.	e) Drafting the proposal document	- Progress check - Discussions and Presentations of what's been done so far - Supervision check	
	Semester Break	27th May – 1st June 2026	Semester Break	27th May – 1st June 2026	
Week 9-10	Proposal Examination (15th June – 3rd July)	At the end of this topic, Students will be able to: a) Resubmit any failed proposal document for re-examination. b) Prepare and design the wireframes of what they expect to use in the project implementation. c) Familiarise themselves with the different tools that they have proposed to use in the project implementation. At the end of this topic, Examiners will be able to: a) Begin the Examination process. b) Relay feedback back to the students for corrections	Assess the presented project designs.	- Progress check - Discussions and Presentations of what's been done so far - Supervision check	
Week 9-13	Git & Github classes	At the end of this topic, Students will be able to: (Content to be covered will be updated)			
Week 13	Proposal Defense (Study Break: 13th – 20th July)	At the end of this topic, Students will be able to: a) Prepare a short presentation (Max 5 slides) for the proposal defence.	Assess the presentations prepared.	- Progress check - Discussions and Presentations of what's been done so far - Supervision check	

		At the end of this topic, Examiners will be required to: - Hand back the marked copies. - Re-examine students whose proposals were rejected previously. These students will be required to resubmit the documents within a week for marking. These students will only score half the given marks. On passing, they will be allowed to proceed to the Implementation stage. - Avail themselves for the Proposal Defences that will be scheduled as part of the examination process.	- Hand back the marked proposal documents to the students. - Follow up any cases of resubmissions that need to be done. - Upload the Proposal Defence Schedule.		
		End of Classes Examinations			
Semester 2: 17th August 2026 to 25th November 2026					
Week	Topic	Intended Learning Outcomes	Hybrid/Hyflex Learning Activities		Assessment
Week 1-7	System Implementation (17th August – 2nd October)	At the end of this topic, students will be able to: - With the guidance of their supervisor, embark on development of the solution. - Divide all the development work into a series of sprints and/or modules. - Plan meetings with the supervisor. The student needs to note down all the meeting that take place with the Supervisor.	Assess project progress	- Progress check - Discussions and Presentations of what's been done so far - Supervision check	

Week 8	Semester Break	7th – 9th October	Semester Break	7th – 9th October	
Week 8-9	Final Documentation (12th – 16th October)	At the end of these weeks, students will be able to: a) Draft the Final Documentation as discussed in class. b) Review the document with the Supervisor. c) Make corrections on the document as advised by the Supervisor. d) Submit the corrected copy for Examination.	Discuss the Final Documentation Guidelines	- Progress check - Discussions and Presentations of what's been done so far - Supervision check	
Week 10-14	Preparation for Final Presentations (19th October – 20th November)	At the end of these weeks, students will be able to: a) Finalize all pending modules/sprints b) Present the final working project c) Prepare a set of slides that will be presented at the defense session.	Discuss the final requirements ahead of the defenses	- Progress check - Discussions and Presentations of what's been done so far - Supervision check	
Week 15	Final Defenses (Study Break: 23rd November – 1st December)	At the end of this topic, students will be able to: a) Avail themselves for the presentations which will be scheduled in advance and the list shared with the students beforehand. b) Prepare a brief ppt presentation (Max 5 slides) and do a project demo before the Panel of Examiners. c) Upload the Final Document for Examination purposes.	- Assess the presentations prepared. - Upload the Final Presentation Schedule.	- Progress check - Discussions and Presentations of what's been done so far - Supervision check	

Course Delivery Methodology

1. Lectures will be used to introduce and discuss various guidelines that will help with the project. Guidelines will be uploaded on E learning.
2. Seminars and Tutorials
3. Consultations with the supervisors

4. Instructional Materials and/or Equipment
5. Audio visual equipment, chalkboard, computer simulation software, computer programming tools

Academic Assessment

Type	Weighting (%)
Final Examination	60
1. Final System Documentation	40%
2. Final Defense	20%
Coursework Examination	40
3. Concept Paper	5%
4. Proposal Document	20%
5. Proposal Defense	15%
Total	100 %

Core Reading Materials

- i. There is no primary text, but various software manuals, research papers and previous project reports, as well as relevant textbooks will be used.
- ii. A range of hardware and software appropriate to the chosen topics; Full use of the University Library, including the Inter-Library Loan facility and all available electronic resources

Assignments/Homework

Students will be given tasks/milestones during the semester, most of which will be part of the semester project which counts as part of coursework evaluation.

Location and Time

Please refer to the official timetable. Lecture rooms may change from time to time and it is important to keep track of communication from the module leader.

Policies

- a) **Punctuality** is fundamental. Active **individual participation** in class presentations is required. It is prohibited to sign the attendance register/log into class on behalf of colleagues who are not present.
- b) **Plagiarism** is a serious offence. If detected in any form in course work and assignments, the following will apply:
 - i. In partial or non-serious cases (such as not citing whole word-for-word quotes), half the total possible marks of the document/chapter are duly struck off.
 - ii. In serious cases (such as whole duplication of a document/chapter), a zero policy will apply i.e., all offending parties will be awarded a mark of zero.
- c) **Note:** The level of seriousness referred to above is at the discretion of the Coordinator/Supervisor/Panel of Examiners. Appeals are certainly possible through the relevant channels.

Communication Channels

- a) E-mail
- b) Course E-Learning Site
- c) Module Leader